



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 5

TITLE: :Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement:

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives:

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory:

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcomes:

ArrayList, Vector, and StringBuffer enable efficient dynamic data storage and string manipulation, allowing flexible addition, removal, and management of data without fixed size constraints.

```
public class JAVA_EXP_5 {  
    public static void main(String[] args) {  
  
        ArrayList<String> arrayList = new ArrayList<>();  
        arrayList.add("Apple");  
        arrayList.add("Banana");  
        System.out.println("ArrayList: " + arrayList);  
  
        Vector<String> vector = new Vector<>();  
        vector.add("Car");  
        vector.add("Bike");  
        System.out.println("Vector: " + vector);  
  
        StringBuffer sb = new StringBuffer();  
        sb.append("Hello");  
        sb.append(" World");  
        sb.insert(5, ",");  
        System.out.println("StringBuffer: " + sb);  
    }  
}
```

```
in JAVA_EXP_5  
ArrayList: [Apple, Banana]  
Vector: [Car, Bike]  
StringBuffer: Hello, World  
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion:

This assignment demonstrates the effective use of ArrayList, Vector, and StringBuffer for dynamic data storage and string manipulation in Java. ArrayList provides fast, resizable storage for single-threaded applications, while Vector offers synchronized, thread-safe operations. StringBuffer enables efficient, mutable string handling without creating multiple objects.

Exercise:

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
import java.util.ArrayList;

public class JAVA_EXP_5{

    ArrayList<String> tasks = new ArrayList<>();

    void addTask(String task){
        tasks.add(task);
    }

    void removeTask(String task){
        tasks.remove(task);
    }

    void displayTasks(){
        StringBuffer sb = new StringBuffer();
        sb.append("To-Do List: \n");
        for (int i = 0; i < tasks.size(); i++){
            sb.append((i+1) + ". " + tasks.get(i) + "\n");
        }
        System.out.println(sb);
    }

    public static void main(String[] args) {
        JAVA_EXP_5 list = new JAVA_EXP_5();

        list.addTask("Buy groceries");
        list.addTask("Complete Java assignment");
        list.addTask("Read a book");

        list.displayTasks();

        list.removeTask("Read a book");
        System.out.println("After removing a task: ");
        list.displayTasks();
    }
}
```

To-Do List:

1. Buy groceries
2. Complete Java assignment
3. Read a book

After removing a task:

To-Do List:

1. Buy groceries
2. Complete Java assignment

PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
import java.util.ArrayList;

public class JAVA_EXP_5{

    ArrayList<String> courses = new ArrayList<>();
    ;

    void addCourse(String course) {
        courses.add(course);
    }

    void removeCourse(String course) {
        courses.remove(course);
    }

    void viewCourses() {
        StringBuffer sb = new StringBuffer();
        sb.append("Registered Courses:\n");
        for (int i = 0; i < courses.size(); i++) {
            sb.append((i + 1) + ". " + courses.get(i) + "\n");
        }
        System.out.println(sb);
    }

    public static void main(String[] args) {
        JAVA_EXP_5 reg = new JAVA_EXP_5();
    }
}
```

```
reg.addCourse("Java Programming");
reg.addCourse("Database Management");
reg.addCourse("Computer Networks");

reg.viewCourses();

reg.removeCourse("Computer Networks");
System.out.println("After removing a course:");
reg.viewCourses();
}
}
```

```
a9029103fc22\redhat.java\jdt_ws\JAVA_PRACTICALS_eabe1187\bin' 'JAVA
Registered Courses:
1. Java Programming
2. Database Management
3. Computer Networks

After removing a course:
Registered Courses:
1. Java Programming
2. Database Management

PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```